

Department for Engineering
Study program Renewable Energy Systems



Students name

Title of the Masters Thesis

Master thesis for obtaining
the academic degree:

Master of Engineering (M. Eng.)

Submission date: DD.MM.YYYY

Matriculation Number 12345

First reviewer: Name of the first reviewer
University of Applied Sciences Nordhausen

Second reviewer: Name of the second reviewer
Organisation

Abstract

On a maximum of half a page, the student should write down a short version (abstract) of the thesis. Based on this abstract, the reader must be able to decide if the thesis contains topics of interest for the reader.

Kurzfassung

Falls angebracht, kann zusätzlich eine Kurzfassung in deutscher Sprache angefügt werden.

Prologue and/or acknowledgement

Preface and/or acknowledgements are optional and need not appear in the paper.

The Master Thesis is the final thesis of the Master programme. The Master programme is finished with the public colloquium. The Master thesis should show that the candidate is able to perform a task in his/her field of expertise within a given period of time and to work independently according to scientific methods and findings. Next to the framework conditions described in this guideline, the conditions and instructions specified in the applicable examination and study regulations obtain regarding composing the thesis.

By means of the content of the thesis is proved that methods of Renewable Energy Systems can be mastered and applied according to the current state of the art. In this context, it is especially important which new ideas and concepts the author has developed. It must also be clear how the author the theoretical background of the work has researched and reviewed.

The outline and structure of the Master Thesis is based on the usual regulations for the preparation of scientific papers. This document contains guidelines for the formal and content-related organisation of Bachelor, Diploma and Master Theses.

(Effective: July 27, 2020)

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Symbol	Unit	Meaning
α	1	solarer Absorptionskoeffizient der oberen Abdeckung
Δp_v	mbar	Druckverlust des PVT-Moduls
ΔT	K	Temperaturdifferenz zwischen T_m und T_a
ϵ	1	IR-Emissionskoeffizient der oberen Abdeckung
η_{hem}	1	Kollektorwirkungsgrad, basierend auf hemisphärischer Globalstrahlung
$\eta_{0,b}$	1	Spitzenwirkungsgrad für Direktstrahlung
γ	$^\circ$	Kollektorausrichtung
β	$^\circ$	Kollektorneigung
σ	$W/(m^2 K^4)$	Stefan-Boltzmann-Konstante
θ	$^\circ$	Einfallswinkel der Direktstrahlung in Kollektorebene
A_{Abs}	m^2	Absorberfläche
A_G	m^2	Bruttokollektorfläche
A_{KF}	m^2	Kollektorfeldfläche
a_1	$W/(m^2 K)$	Kollektorparameter (KP) für ΔT -Abhängigkeit
a_2	$W/(m^2 K^2)$	KP für ΔT^2 -Abhängigkeit
a_3	$(Ws)/(m^2 mK)$	KP für ΔT - und Windabhängigkeit
a_4	1	KP für IR-Strahlungsabhängigkeit
a_5	$kJ/(m^2 K)$	Kollektorparameter, entspricht der effekt. Wärmekapazität
a_6	s/m	KP für Windabhängigkeit u und Einstrahlung G
a_7	s/m	KP für IR-Strahlungs- und Windabhängigkeit
a_8	$W/(m^2 K^4)$	KP für ΔT^4 -Abhängigkeit
b_1	$W/(m^2 K)$	Kollektorparameter für ΔT -Abhängigkeit
b_2	$Ws/(m^3 K)$	Kollektorparameter für ΔT - und Windabhängigkeit
b_u	s/m	Kollektorparameter für Windabhängigkeit von $\eta_{0,hem}$
b_0	1	Kollektorparameter zur Einfallswinkelkorrektur ($K_b(\theta)$)
c_{eff}	$kJ/(m^2 K)$	effektive Wärmekapazität des PVT-Moduls einschl. Fluid
c_T	$\%/K$	Leistungstemperaturkoeffizient des PV-Moduls
E_L	W/m^2	langwellige Einstrahlung (IR) in Kollektorebene
G	W/m^2	hemisphärische solare Bestrahlungsstärke (Globalstrahlung in Kollektorebene)
G''	W/m^2	Nettobestrahlungsstärke in Kollektorebene
G_b	W/m^2	Direkte solare Bestrahlungsstärke in Kollektorebene
G_d	W/m^2	Diffuse solare Bestrahlungsstärke in Kollektorebene
IR	-	Infrarotstrahlung
$K_b(\theta)$	1	Einstrahlwinkelkorrekturfaktor für Direktstrahlung

Symbol	Unit	Meaning
K_d	1	Einstrahlwinkelkorrekturfaktor für Diffusstrahlung
$K_{hem}(\theta)$	1	Einstrahlwinkelkorrekturfaktor für hemisph. Globalstrahlung
KI	-	Konfidenzintervall
KP	-	Kollektorparameter
MAE	-	mean absolut error
MBE	-	mean bias error
$\dot{m}$	kg/h	Wärmeträger-Massenstrom
$P_{el,m}$	W	gemessene elektrische Modulleistung
PVT	-	Photovoltaisch - Thermischer Kollektor
$\dot{Q}$	W	Wärmeleistung des Kollektors
Q_{ACO}	kWh/a	Annual Collector Output (dt.: Bruttowärmeertrag)
QDT	-	Quasi Dynamic Testing - Prüfverfahren nach DIN EN ISO 9806
$\dot{Q}_m$	W	gemessene Wärmeleistung
$\dot{Q}_s$	W	simulierte (mit Modell berechnete) Wärmeleistung
$\dot{Q}_{peak}$	W	Spitzenleistung des Moduls
r	1	Kollektorparameter zur Einfallswinkelkorrektur ($K_b(\theta)$)
RMSE		root mean square error
SRC	-	Standardberichtsbedingungen nach DIN EN ISO 9806
SST	-	Steady State Testing - Prüfverfahren nach DIN EN ISO 9806
STC	-	Standardbedingungen zur Angabe der elektr. Nennleistung
T	-	T-Verhältnis, Quotient aus Mittelwert und Standardabweichung
T_a	$^{\circ}C$	Umgebungstemperatur
u	m/s	Windgeschwindigkeit in der Kollektorebene
U_{int}	$W/(mK)$	Interner thermischer Leitwert des Absorbers
$\dot{V}$	l/h	Volumenstrom des Wärmeträgerfluids
WISC		Wind and / or Infrared Sensitive Collector

List of abbreviations

In the text, abbreviations should be used as few as possible. Only regular abbreviations (e.g. from Duden) such as: etc., etc., are recognized.

Abbreviations for convenience are not allowed. Abbreviations used in specialist field (factual type) can be used; however, they are to be listed in the list of abbreviations

in.RET Institut für Regenerative Energietechnik

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Chapter 1

Introduction

It is recommended that the introduction has a length of two to three pages. It helps the reader to get into the subject, familiarize with the task or problem and classifies the topic into a major context. In particular, the motivation for the topic is to be displayed.

1.1 Task

Text...

1.2 Overview

After illustrating the objective of the thesis, the solution approaches and the relations between particular chapters should be shown.

Chapter 2

Basics

The second chapter of a scientific paper is devoted to the basics, which a professionally trained readers should know in order to comprehend all the following chapters.

This fictional reader has a basic engineering education in your field and therefore has the complete basic knowledge. It is therefore sufficient to provide him with the specialized knowledge (e.g. the content of product standards used) required to understand the work.

Chapter 3

Main part A, e.g. Measurement concept

The main part (choose a content-related heading!) of the thesis contains the complete detailed description of the problem solving. An essential aspect for the composition is the effective transfer of the written information to a skilled reader who is not immediately familiar with the problem dealt with. This includes:

- a clear representation of the chain of thoughts, which leads the reader from the state of knowledge at the beginning of the work towards the developed solution at the end of the thesis
- a good structure and logical arrangement of the essential topics
- precise, understandable, short and not verbose wording

Based on the results of preliminary examinations and previous publications the student has the knowledge to write a thesis. In order to solve a problem, several approaches should be discussed - the chosen solution must be explained and defined. Also investigations, which are regarded as essential in terms of scope, but not directly for problem solving should be stated with explanation.

The structure of the topics and thus the order of exposition must orientate to objective aspects and not according to the temporal chronology of the development. The section headings should be chosen as meaningful as possible.

With the help of the description of the conducted researches, the achieved results must be reproducible.

If results or solutions are adopted from another credible source (with reference source), the student has to explain the results so far as necessary for the understanding; in this case, a derivation can be omitted.

Note: The main part usually consists of several chapters!

Starting on the second page, the name of the chapter appears in the header. There the page number should also be found.

Chapter 4

Main part B, e.g. Measurements

Chapter 5

Main part C, e.g. Evaluation

Chapter 6

Main part D, e.g. Results

Chapter 7

Summary (and Outlook)

The work is concluded with a "‘summary'", which can also be a "‘summary and outlook'" with regard to unresolved problems and further need for research.

The summary as the last section of the thesis contains clear statements about:

- the procedures used for processing
- the results of the thesis and their meanings
- the limits of validity and the progresses towards the knowledge at start of thesis
- the applicability of the results

The summary is suitable to emphasize the own contribution within the thesis.

A reader who is looking for information on a particular topic should get, after reviewing the summary and introduction, a good overview of the most important topics of the thesis. Therefore, in the summary statements may be repeated, which have already been described in detail in the main part of the work.

Appendix A

Notes on organisational procedures

A.1 Purpose of the Annex

In the appendix you will find all figures/ tables/ derivatives/ program source texts etc. which are not necessary for immediate understanding, but which must nevertheless be consulted in order to understand the work.

Documents of an exclusively documentary character (e.g. raw data of the measurement acquisition) are to be filed in the electronic appendix in a CD inserted on the back binding.

A.2 Information about the M860 Master Thesis Module

The total workload for the Master Thesis module Module is 900 hours; this corresponds to 30 ECTS credits. This workload results mainly from the independent and self-responsible handling of the master thesis project (700 hours), the preparation of the master thesis (100 hours) and the presentation (100 hours).

A.2.1 A: Master Thesis

With the master thesis the students develop an independent scientific work. The results are usually derived from the practical activity to a given topic in a company or in another enterprise of professional scientific experience.

A.2.2 B: Presentation and Defence (Colloquium)

In the Colloquium, students should present the results of their master thesis in an oral presentation and answer the questions of the supervisors and the audience.

A.2.3 Learning goals

By completing the master thesis module, the students prove their professional qualification, to solve a practice-relevant engineering problem within a given time limit independently and with scientific methods, to present the results succinctly in a professional manner and to defend it in a colloquium.

A.3 General Notes

The following paragraph is an extract from the Examination regulations for the Master's programmes:

A.3.1 Purpose and conduct of the Master's examination

1. The Master's examination forms the conclusion of the Master's programme. The Master's examination determines whether the candidate has an overview of the interrelationships of his subject, has the ability to apply scientific methods and findings, in particular, and has acquired the necessary indepth specialist knowledge.
2. As a rule, the examination results of the Master's examination must be completed after the respective courses during the course of study. The Master's examination is concluded with the Master's thesis and the Master's colloquium.

A.3.2 Type and Scope of the Master's Examination

1. The Master's examination consists of the examination results, the Master's thesis and the Master's colloquium.
2. Examinations must be taken in accordance with the study regulations. The subject of the examination papers are the subject areas of the study units or modules assigned to them in accordance with the study regulations. The type of examination performance in the individual modules is generally specified in the study regulations.
3. The candidate receives ECTS credits in accordance with the study regulations for passed examinations.

A.3.3 Output and Processing Time of the Master Thesis

1. The Master's thesis should show that the candidate is able to work independently and with scientific methods on a practically relevant problem from his subject within a given period of time.
2. The topic of the Master's thesis is presented by a professor or another person entitled to sit the examination in accordance with § 17 of these regulations and issued by the responsible examination board.
3. The topic of the Master's thesis may only be spent if 45 ECTS credits have been earned.

4. The procedure for issuing the Master's thesis is regulated by the examination board. The subject and time shall be recorded. The candidate may express requests for topics. In justified cases, the topic can be returned once and only within four weeks of issue.
5. The processing time for the Master's thesis is a maximum of five months. The topic, task and scope of the Master's thesis must be limited in such a way that the deadline for processing the Master's thesis can be met. The processing time may be extended by up to four weeks at the candidate's request for reasons beyond his control.

A.3.4 Submission, Evaluation and Repetition of the Master's Thesis

1. The Master's thesis must be submitted to the Studien-Service-Zentrum (SSZ) in bound form and in triplicate by the deadline; the time of submission must be recorded on a record. With the approval of the first examiner of the university, the Master's thesis can also be submitted in another form. In any case, a bound copy must be submitted for archiving. At the time of submission, the candidate must declare in writing in an attached declaration that he/she has written his/her work independently and has not used any other sources and aids than those stated. This declaration must be attached to the copies submitted.
2. A Master's thesis that is not submitted on time according to paragraph 1 must be evaluated as "insufficient" (5.0).
3. The review and evaluation of the Master's thesis is carried out by a first examiner and a second examiner. The grade of the Master's thesis is calculated from the average of the individual grades of the examiners in the case of grade differences. If the examiners' assessments differ by more than two marks or if one of the examiners awards the mark "insufficient" (5.0), a third examiner must be included. In this case, the overall grade is the arithmetic mean of all three evaluations.
4. The Master's thesis can only be repeated once in the case of an assessment that is worse than "sufficient" (4.0). The topic of the Master's thesis may only be returned within the period specified in § 21 Paragraph 4 Sentence 4 if the candidate has not made use of this option when writing his first failed Master's thesis.
5. The review and evaluation of the Master's thesis must be completed after three months at the latest.

A.3.5 Master Colloquium

1. The Master's Colloquium is the professional conclusion of the course of study. The candidate should show that he is able to analyse and work on practical problems from his field systematically, independently and according to scientific methods.
2. A colloquium will give the candidate the opportunity to present and defend his master's thesis. The colloquium is limited to questions about the course of study and the subject area from which the Master's thesis is taken.
3. The Master's Colloquium is usually held before the first examiner of the Master's thesis and is attended by the second examiner of the Master's thesis. The duration is usually 60 minutes. A Master's colloquium that has not been passed can be repeated once.
4. The Master's Colloquium can only be scheduled if proof of all examination achievements of the Master's examination has been furnished and the Master's thesis has been assessed as at least sufficient.
5. The main topics and results of the Master's Colloquium must be recorded in a protocol and signed by the examiner and the observer. The result shall be communicated to the candidate on the same day.

Appendix B

Notes on Formal Layout

B.1 Formalities

B.1.1 Scope, Pattern, Binding and CD-Rom

The usual size of a Master Thesis amounts to about 80 pages¹ in the format A4. The text is to be created in single-column, full justification and one-sided with 1.5-fold line spacing. The candidate can choose between the fonts: News Gothic MT 12 pt, Times New Roman 12 pt or Arial 12 pt.

For all text pages, even those with inserted illustrations, tables and sheets, the edge widths of 2.0 cm (below) and 2.5 cm (left, right, top) must be observed.

Master theses has to be bound and provided with a hard cover. Ring binding will not be accepted. The cover must be bound with a cardboard binding (glued or tied). The imprint on the back of the cover should show the name of the student as well as the year of creation.

The CD-ROM (electronic appendix) contains the Master Thesis in electronic form (preferably in PDF and Word format) and all information used that is not documented in the appendix. This can be e.g. xls-files with measurement data, calculations or pdf-copies of the used web sources. For the CD-ROM, a CD cover must be provided at the back cover.

¹Theses, which contains elaborate measurements, usually contains more than 100 pages because the description of measurement setup, sensors, data analysis, etc. often uses many pages. It rarely does succeed, to create a Master thesis with only 60 or fewer pages.

B.1.2 Table of Contents and Numeration

The table of contents lists the numbered section headings and page numbers of the beginning of each section. The numbering of the sections should begin with 1. A decimal point following the last number of the numeration is not to be used (2.2.2 is correct, 2.2.2. is wrong).

Headings in text and table of contents must match exactly. Headlines of particular outline items should not cover the overall topic of the thesis and / or the headings of sub-items.

A maximum of three outline levels should be used (2, 2.1, 2.1.1, but not 2.1.1.1). If one point should be followed by a sub-item, then the minimum number of these sub-items are two (for example, if there is a point 2.2.1, there must be at least one point 2.2.2).

Each main section (1, 2, etc.) starts on a new page. Text pages in the appendix are numbered separately by subjecting to the appendix number.

B.1.3 Figures, Charts etc.

Figures and tables must have a clear content description. It is very important to pay attention to an accurate local, temporal and factual distinction of the representation. The figures and tables are to be numbered consecutively, with table headings and image captions.

Example for the integration of a table

Table B.1: Example

feature	1	2	3
A			
B			

A short explanation of the symbols/abbreviations used (without prejudice to their detailed explanation in the text) shall be placed immediately below tables or flow charts/schemes.

Example for the integration of a figure

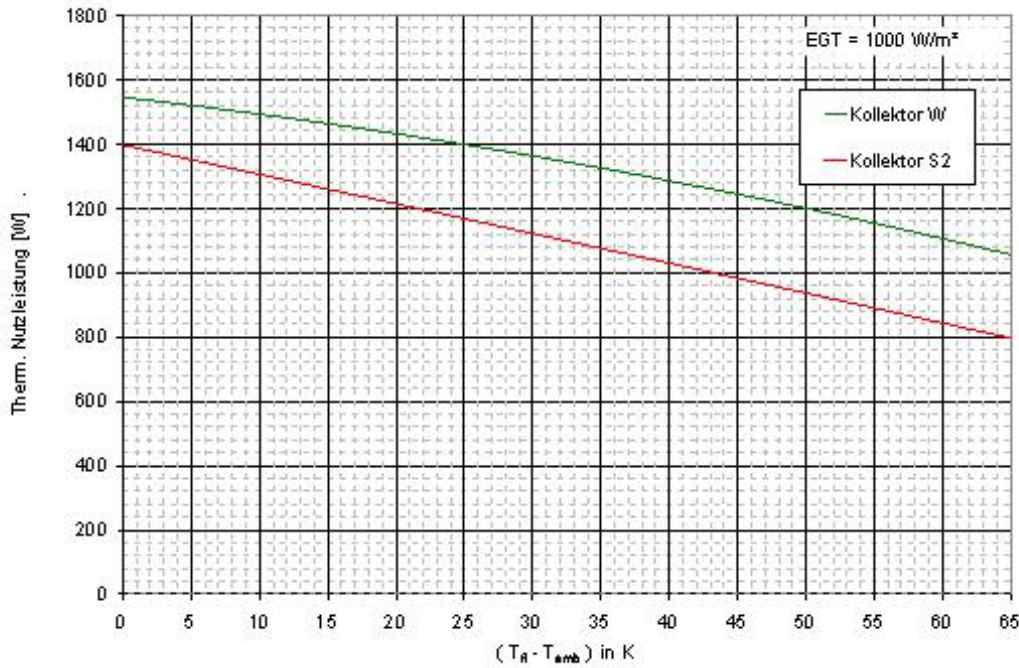


Figure B.1: Example figure. The text in the figures must be easy to read, this means the student should use the same font sizes in illustration and flow text; in this example, the readability is not given. Tip: Insert figures from Excel via *Edit - Insert contents - graphic*, do not insert as a Windows object.

The space between text and figures etc. should be the same. There must be a reference in the text for all figures / tables / formulas / attachments, etc.

Example for the integration of an equation or formula

Even formulas or equations, referred to in the text, must be continuous numbered. It is a good practice to use the number of the main chapter in the first place for numeration and a continuous number after a dot for the equations, figures or tables in this main chapter.

$$F = m \cdot a \quad (\text{B.1})$$

Example for the application of a multiline equation:

$$\begin{aligned}
 \dot{Q}/A_G &= \eta_{0b} \cdot K_b(\theta) \cdot G_b + \eta_{0b} \cdot K_d \cdot G_d \\
 &- a_1 \cdot \Delta T - a_2 \cdot \Delta T^2 - a_3 \cdot u \cdot \Delta T \\
 &+ a_4 \cdot \sigma \cdot (T_{sky}^4 - T_a^4) - a_5 \cdot \frac{\partial T_m}{\partial t} \\
 &- a_6 \cdot u \cdot G - a_7 \cdot u \cdot \sigma \cdot (T_{sky}^4 - T_a^4) - a_8 \cdot \Delta T^4,
 \end{aligned} \quad (\text{B.2})$$

B.1.4 References

References in the text to other sections of the text are preferably given by specifying the page number (not the section number). References to figures, tables or formulas are made by specifying the corresponding number and page number.

For Cross-references in the text to a reference, e.g. figure (Fig. B.1), formula (eq. B.1) or table (Tab. B.1) use in WORD:

Insert - Reference - Cross reference - Selection of the reference type

B.1.5 Abbreviations and symbols

In the text, abbreviations should be used as few as possible. Only regular abbreviations (e.g. from Duden) such as: etc., etc., are recognized. Abbreviations for convenience are not allowed. Abbreviations used in specialist field (factual type) can be used; however, they are to be listed in the list of abbreviations.

The thesis should use a uniform symbolism. Does the student uses symbols from foreign sources, the symbols have to be adapted to those symbols being used in the thesis. Exceptions to this are literal quotations.

B.1.6 Spelling and Language

Basically, the student has to write the Master thesis in German or English language. The student should use British or American English. In coordination with the supervisor, exceptions are possible.

B.1.7 Citation guidelines

Scientific work is characterized that the author has to work with clear references. This allows the reader to refer to the cited sources and thus to verify the origin of all scientific statements, as far as they are not self-developed ideas.

The student need to quote all the phrases, sections, as well as unaltered illustrations etc. that literally are taken from other publications. Collection of quotations should be avoided. The student need to identify every section, example, modified Illustration and other sources that have been copied with complete references.

There are many recommendations how a list of references can be designed. It is important that the list of references is consistent. Authors, titles of publications or details about publishers should be formatted uniformly.

Below, required information for various publications are compiled in accordance with [1]. The references [2] to [5] in the list of references of this document act as examples of entries from:

- *books* (see entry [2]),
- *Journals* (see entry [3]),
- *Conference proceedings* (see entry [4]) und
- dem *Internet* (see entry [5]).

Instead of the citation method selected here (numerically in square brackets), the student can e.g. also quote alphanumerically (combination of abbreviated author and year in square brackets). The list of references can be numerically (according to citation order in the text) and / or alphabetically sorted.

The structure of the list of references should be based on the following scheme:

Quotes from Books

For quotes from books, the following rule applies:

Name1, Firstname1; Name2, Firstname2; ...: *Title : Subtitle*. Attributes of edition, (short) designation of the publisher, place of publication (year of publication), p. xx - yy.

Note: dot behind the title, dot in front of the page reference. Titles and subtitles, if present, are separated by space – double dot – space. For better emphasizing, the title is in italic. If there are more than two authors you may continue with “et al.” after the first author.

Quotes from Journals

For journals, the following rule applies:

Name1, Firstname1; Name2, Firstname2; ...: Title : Subtitle. In: *title of journal* **reel- or vintage number** (year of publication) journal number, p. xx - yy

Note: For better emphasizing, the title of journal is in italic.

Quotes from Conference Proceedings

For quotes from *conference proceedings*, the following rule applies:

Name1, Firstname1; Name2, Firstname2; ...: *Title : Subtitle*. In: Conference proceeding, publisher, Designation of the conference, date, p.r.n. place and expert association, p. xx - yy..

Note: For better emphasizing, the title is in italic.

Quotes from Internet

The student should only use the internet as a source if there are no printed publications about the topic available.

For articles from the internet, the following rule applies:

Organisation / Manufacturer / Author: Title: Subtitles. URL (date of accessing: dd.mm.yyyy)

Note: For better emphasizing, the title is in italic. Because internet references are subject to a high conversion, the quoted URLs should be as stable as possible. Make a PDF-copy from the web source for the electronic appendix (see section B.1.1, 14.

Bibliography

- [1] Norm DIN 1505 Teil 2: *Titelangaben von Dokumenten : Zitierregeln* (1984).
- [2] Gerthsen, C.: *Physik*. Springer, Berlin (1995).
- [3] Hammer, A., Heinemann, D., Hoyer, C., Kuhlemann, R., Lorenz, E., Müller, R.W., Beyer, H.G.: Solar Energy Assessment Using Remote Sensing Technologies. In: *Remote Sensing of Environment* **86** (2003) 3, S. 423–432.
- [4] Meyer, R., Butron, J.T., Marquardt, G., Schwandt, G., Geuder, N. Hoyer-Klick, C., Lorenz, E., Hammer, A. Beyer, H.G.: *Combining Solar Irradiance Measurements and Various Satellite-Derived Products to a Site-Specific Best Estimate*. In: Proceedings of the 14th biennial SolarPACES Symposium, 04.03.–07.03.2008, Las Vegas, USA.
- [5] Hochschule Nordhausen: *Prüfungsordnung für die Bachelorstudiengänge des Fachbereichs Ingenieurwissenschaften an der Hochschule Nordhausen*. Amtliche Bekanntmachungen der HSN, Nr. 11/2018. <https://www.hs-nordhausen.de> (Abruf: 13.12.2019)

Statutory Declaration

Each Master Thesis must contain a statutory declaration signed by the student, which says that the work was written independently. Wording as follows:

"I herewith declare that I have composed the present thesis myself and without use of any other than the cited sources and aids. Sentences or parts of sentences quoted literally are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned.

In particular, I have not utilized commercial advice. Third parties have neither received direct nor indirect monetary benefits from me for actions which are related to the content of the submitted work.

The thesis in the same or similar form has not been submitted to any examination board and has not been published. This thesis was not yet, even in part, used in another examination or as a course performance, either in Germany or abroad."

Nordhausen, TT.MM.JJJJ

Complete handwritten signature